

O2C ORDER MANAGEMENT SYSTEM

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1. Project Overview

This project implements a simplified Order-to-Cash (O2C) business process using the SAP ABAP RESTful Application Programming Model (RAP). The system handles customer and order data, integrates backend logic with modern UI exposure, and demonstrates enterprise-grade architecture design using CDS Views and OData services.

2. Data Model Design

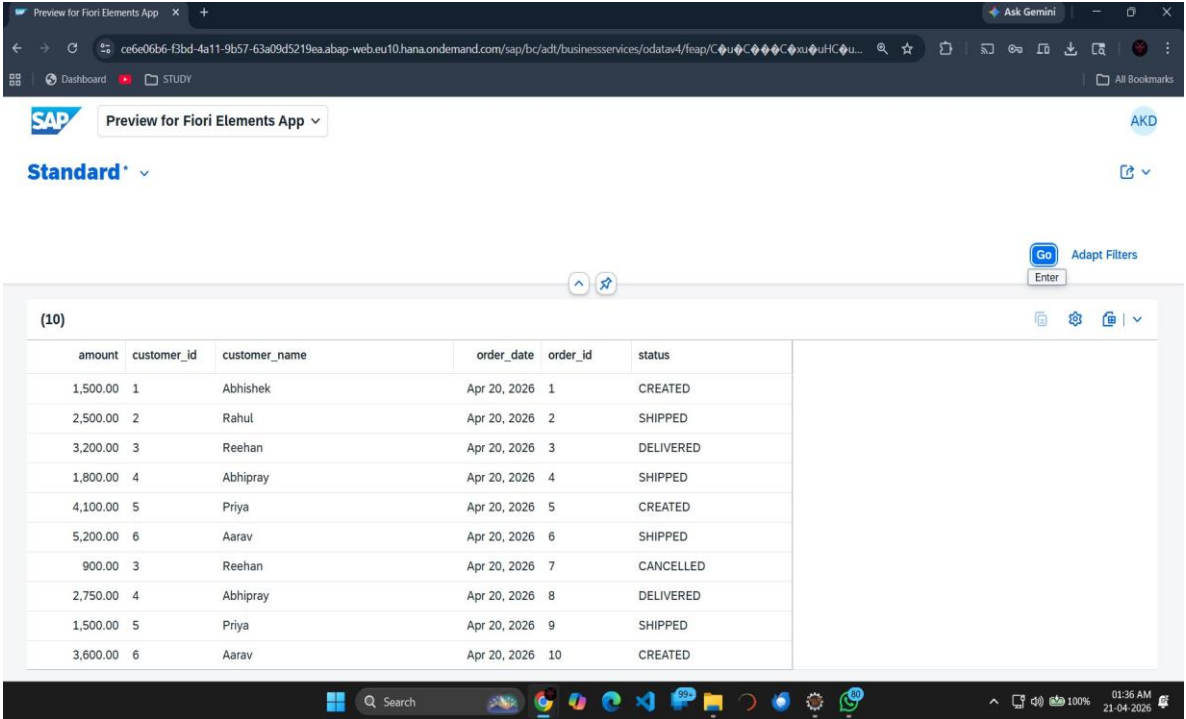
ZO2C_CUSTOMER and **ZO2C_ORDER** are the core database tables storing master and transactional data respectively. The design ensures normalization, scalability, and real-world mapping of customer-order relationships.

3. Core Data Services (CDS)

ZO2C_ORDER_ROOT defines the root entity capturing order attributes like Order ID, Customer ID, Date, Amount, and Status.

ZO2C_REPORT_VIEW performs an inner join between customer and order tables to provide enriched reporting data with customer names.

4. Implementation & Code Explanation

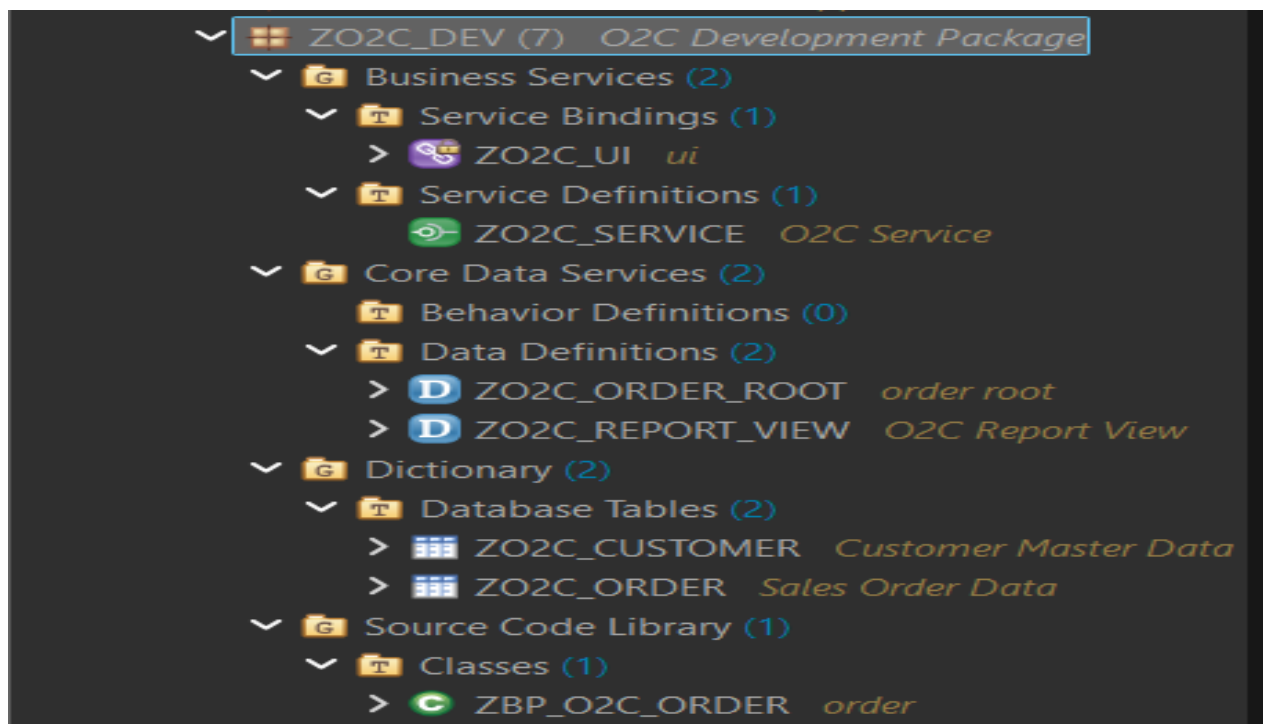


The screenshot displays the SAP Fiori Elements App interface. At the top, there's a navigation bar with the SAP logo and a dropdown menu labeled 'Preview for Fiori Elements App'. Below this, the word 'Standard' is visible. The main content area shows a table with 10 rows of data. The table has columns for 'amount', 'customer_id', 'customer_name', 'order_date', 'order_id', and 'status'. The data is as follows:

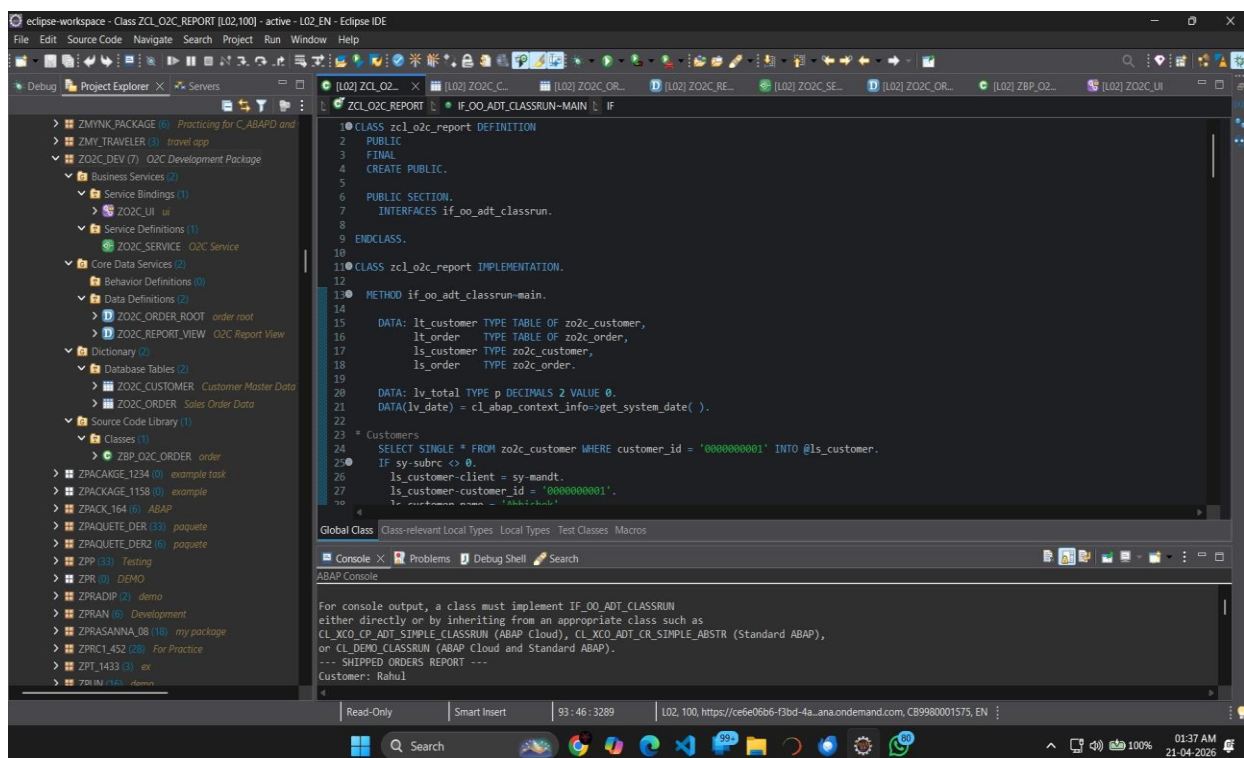
amount	customer_id	customer_name	order_date	order_id	status
1,500.00	1	Abhishek	Apr 20, 2026	1	CREATED
2,500.00	2	Rahul	Apr 20, 2026	2	SHIPPED
3,200.00	3	Reehan	Apr 20, 2026	3	DELIVERED
1,800.00	4	Abhipray	Apr 20, 2026	4	SHIPPED
4,100.00	5	Priya	Apr 20, 2026	5	CREATED
5,200.00	6	Aarav	Apr 20, 2026	6	SHIPPED
900.00	3	Reehan	Apr 20, 2026	7	CANCELLED
2,750.00	4	Abhipray	Apr 20, 2026	8	DELIVERED
1,500.00	5	Priya	Apr 20, 2026	9	SHIPPED
3,600.00	6	Aarav	Apr 20, 2026	10	CREATED

The table is displayed in a light blue theme. At the bottom of the screenshot, the Windows taskbar is visible, showing the search bar, task view button, and several application icons. The system clock indicates 01:36 AM on 21-04-2025.

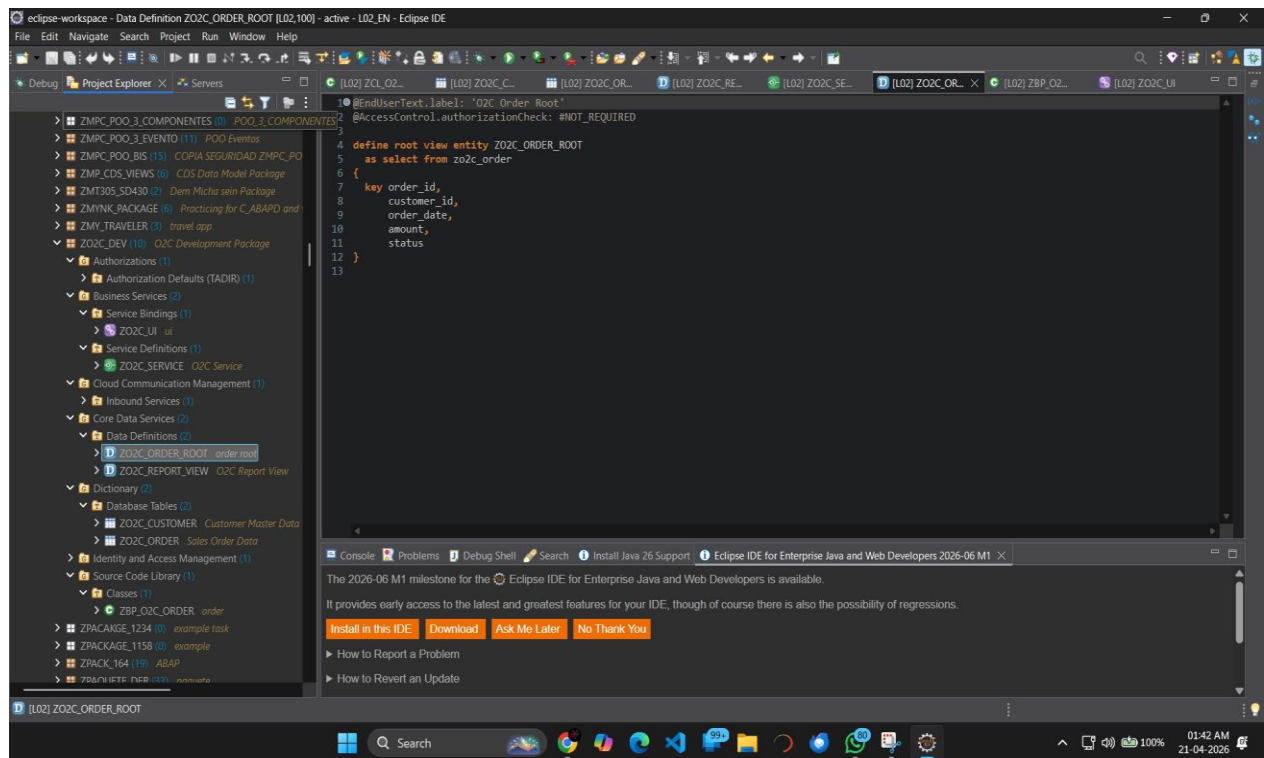
Explanation: Project structure showing RAP objects and organization. This part of the implementation ensures proper data flow between database, CDS layer, and service exposure, forming the backbone of the RAP model.



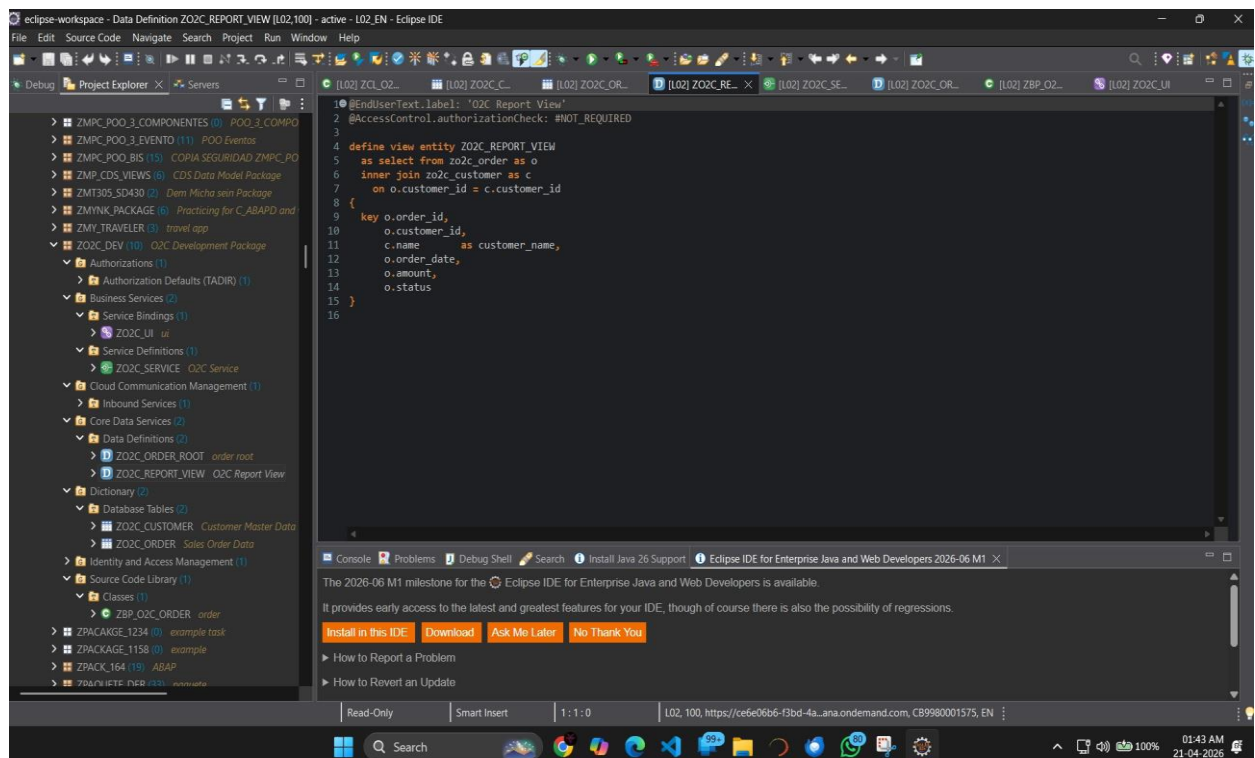
Explanation: ABAP Class implementing business logic and test execution. This part of the implementation ensures proper data flow between database, CDS layer, and service exposure, forming the backbone of the RAP model.



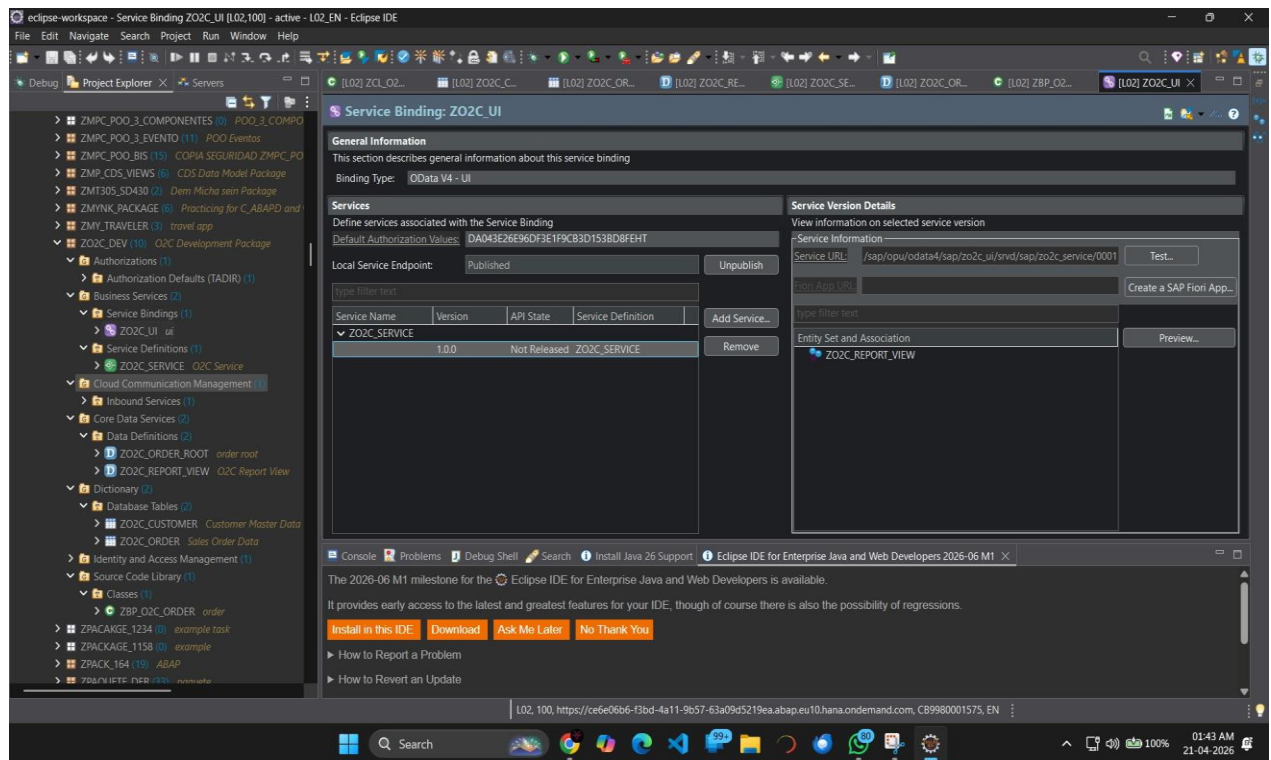
Explanation: CDS Root View representing base order entity. This part of the implementation ensures proper data flow between database, CDS layer, and service exposure, forming the backbone of the RAP model.



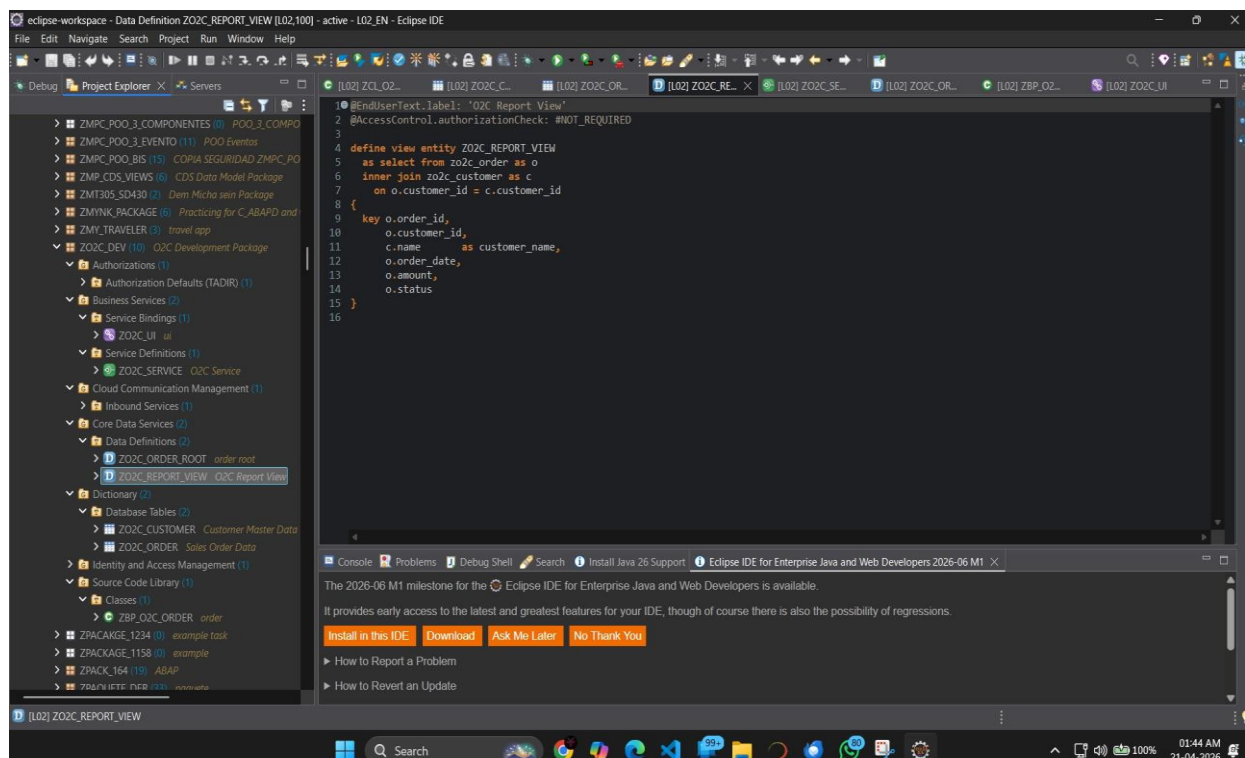
Explanation: CDS Report View combining customer and order data. This part of the implementation ensures proper data flow between database, CDS layer, and service exposure, forming the backbone of the RAP model.



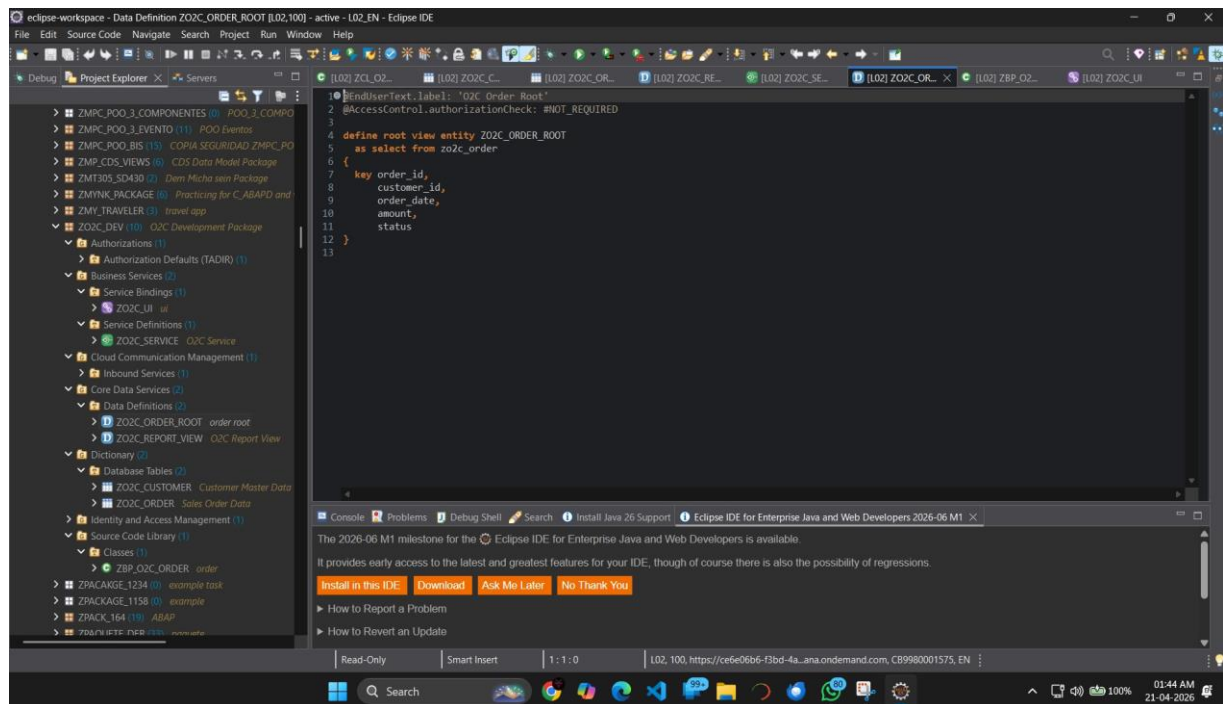
Explanation: Service Binding exposing CDS as OData service. This part of the implementation ensures proper data flow between database, CDS layer, and service exposure, forming the backbone of the RAP model.



Explanation: CDS join logic and field mapping. This part of the implementation ensures proper data flow between database, CDS layer, and service exposure, forming the backbone of the RAP model.



Explanation: Root entity structure definition. This part of the implementation ensures proper data flow between database, CDS layer, and service exposure, forming the backbone of the RAP model.



Explanation: Service Binding preview and entity exposure. This part of the implementation ensures proper data flow between database, CDS layer, and service exposure, forming the backbone of the RAP model.

5. Service Layer (OData V4)

The Service Definition and Service Binding expose the CDS View as an OData V4 service. This enables seamless integration with SAP Fiori Elements UI, allowing real-time data visualization and interaction.

6. User Interface (Fiori Elements)

The application is rendered using Fiori Elements, automatically generating a responsive UI based on metadata. It displays order records with statuses such as CREATED, SHIPPED, DELIVERED, and CANCELLED.

7. Output & Results

The final output successfully displays multiple customer orders with proper relational mapping. The UI is clean, scalable, and aligned with SAP enterprise standards.

8. Conclusion

This project demonstrates a complete RAP-based application lifecycle—from data modeling to UI exposure. It highlights the power of modern ABAP development in building scalable enterprise applications.